

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

November 2013

CL 404 Water Resources Engineering & Management

Date: 21.11.2013, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

## SECTION – I

- Q – 1 (a) Distinguish between the following: [03]  
(i) Drought and Flood (ii) Design flood and Maximum probable flood
- (b) Explain the importance of the rivers in the development human civilization. Also, [04]  
distinguish between perennial river and virgin river.
- Q – 2 (a) What are the different types of investigation necessary for the planning of a reservoir? [05]  
Explain the importance of any ONE of them.
- (b) Discuss, with a suitable sketch, different zones of storage of the reservoir. What is the [05]  
significance of the dead storage?
- (c) What is river training works? What are the objectives of river training works? [04]

## OR

- Q – 2 (a) Differentiate clearly between mass curve and demand curve. [03]
- (b) Enlist different types of river training works. Discuss, in detail, any ONE of them. [04]
- (c) A contour survey of a reservoir site gives the following data: [07]

| Contour value    | Area          |
|------------------|---------------|
| At 200 m contour | 6.0 hectares  |
| At 210 m contour | 18.1 hectares |
| At 220 m contour | 34.0 hectares |

The capacity of the reservoir upto 200 m elevations is found to be 14.1 ha.m  
Determine the general equation for the area-elevation curve and capacity-elevation curve. Also determine the reservoir capacity at RL 230 m.

- Q – 3 (a) What is 'surge tank'? Explain its importance in the safety of the power plant. [04]
- (b) Enlist different structural measures of flood control. What are the favorable [04]

conditions for the floodways?

- (c) Draw a neat sketch showing different components of a hydro electric project. Discuss the functions of each of them. [06]

OR

- Q - 3 (a) Explain the terms: [04]  
(i) Plant factor (ii) Installed capacity (iii) Firm power (iv) Load factor  
(b) Discuss various non-structural measures for the flood control. [04]  
(c) Discuss the advantages and disadvantages of hydro-power plants as compared to other power plants. [06]

SECTION - II

Q - 4 Answer the followings:

1. Sedimentation of a reservoir can be reduced to some extent by venting out the density currents. True or False? [01]
2. The firm yield is usually taken as the design yield. True or False? [01]
3. Enlist steps involved in planning of a WRDP. [01]
5. Define: Catchment yield, Reservoir yield, Physical independent unit, Economically independent unit [04]

- Q - 5 (a) What are physical factors affecting WRDP? [03]  
(b) Explain how the environmental effects influenced the planning of a resource project. [04]  
(c) The daily flows in a river for two years are listed in the table given below. Compute the annual and daily flow for 70% dependability of river flow. [07]

| Sr. No. | Mean Value of Daily Discharge, (m <sup>3</sup> /sec) | Year 1998            | Year 1999            |
|---------|------------------------------------------------------|----------------------|----------------------|
|         |                                                      | No. of days the flow | No. of days the flow |
| 1       | 95.05                                                | 0                    | 6                    |
| 2       | 85.05                                                | 16                   | 19                   |
| 3       | 75.05                                                | 27                   | 25                   |
| 4       | 65.05                                                | 21                   | 60                   |
| 5       | 55.05                                                | 43                   | 51                   |
| 6       | 45.05                                                | 59                   | 38                   |
| 7       | 35.05                                                | 64                   | 29                   |
| 8       | 25.05                                                | 22                   | 48                   |
| 9       | 15.05                                                | 59                   | 63                   |
| 10      | 5.05                                                 | 54                   | 26                   |

OR

- Q - 5 (a) Enlist different types of losses in reservoir. [02]



- (b) How can the sedimentation in reservoir be controlled? [05]
- (c) Which curves are generally plotted for the economic analysis of WRP? Draw each of them and explain how the decision about the size of project can be made? [07]

Q - 6 (a) The runoff from a catchment area during successive months in a year is given below. [07]

Determine the maximum capacity of the reservoir required if the entire volume of water is to be drawn off at a uniform rate, without any loss of water over the spillway.

Use Graphical method only.

| Month                     | Jan | Feb | Mar  | April | May  | June | July | Aug  | Sept | Oct  | Nov  | Dec  |
|---------------------------|-----|-----|------|-------|------|------|------|------|------|------|------|------|
| Runoff (Mm <sup>3</sup> ) | 1.3 | 2.0 | 2.70 | 8.50  | 12.0 | 12.0 | 19.0 | 22.0 | 2.50 | 2.20 | 1.90 | 1.70 |

- (b) A proposed reservoir has a capacity of 500 ha.m and catchment area 200 km<sup>2</sup>. The annual stream flow averages 11 cm of runoff and production of annual sediment is 0.040 ha.m / km<sup>2</sup>. Calculate the annual sediment inflow in the reservoir in Mm<sup>3</sup>. [04]
- (c) For better planning and management of project which strategies are carried out? [03]

**OR**

Q - 6 (a) A multipurpose reservoir is designed for irrigation, hydropower and flood control. [08]

The total estimated cost of the project is Rs. 20 crores. The separable costs, estimated benefits and the cost of the alternate single-purpose projects in crores of rupees are given in the table below. Determine the allocation of cost to each of the three functions by (a) Alternative justifiable expenditure method, (b) Remaining benefits method. All the costs are in crores of rupees.

| Row No. | Item                                  | Flood Control | Hydropower | Irrigation | Total |
|---------|---------------------------------------|---------------|------------|------------|-------|
| 1       | Separable costs                       | 3.0           | 6.5        | 5.5        | 15.0  |
| 2       | Estimated benefits                    | 4.0           | 12.5       | 9.5        | 25.5  |
| 3       | Alternate single-purpose project cost | 4.5           | 9.0        | 8.5        | 22.0  |

- (b) What are the needs of Water Resources Planning? Discuss in brief. [03]
- (c) Explain the following terms: [03]
- (i) Density Currents (ii) Trap Efficiency (iii) Capacity Inflow Ratio

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